

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0767

Loudon 345 kV
April 23, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0767. The Interconnection Customer proposes a 253.7 MW wind facility interconnecting at the Loudon 345 kV substation in the Virginia area. The request was received on March 27, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 342 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$46,110,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Indian Lake / Albert Pike 138kV	Thomaston / Loudon 345kV	155.8	307	45.8
Yardley Ridge / Brambleton 138kV	Loudon / Thomaston 345kV Line	141.7	372	45.8
Brambleton / Albert Pike 138kV	Louon / Thomaston 345kV	137.7	358	45.8
Claytonia / Pleasant View 138kV	Loudon / Thomaston 345kV line	136.8	221	45.8
Loudon / Yardley Ridge 138kV	Thomaston / Loudon 345kV	122.4	534	64.2
Pleasant View / Indian Lake 138kV	Loudon / Thomaston 345kV	122.2	288	45.8
Ludon 345-138 kV Transformer	Loudon / Thomaston 345kV	111.0	882	99.4
Jet / Poland Rd 138kV	Loudon / Thomaston 345kV Line	102.8	207	32.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Albert Pike / Indian Lake 138kV	Rebuild line with higher-capacity conductor	\$4,250,000	100% (\$4,250,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yardley Ridge / Brambleton 138kV	Rebuild line with higher-capacity conductor	\$1,983,000	100% (\$1,983,000)
Brambleton / Albert Pike 138kV	Rebuild line with higher-capacity conductor	\$3,225,000	100% (\$3,225,000)
Claytonia / Pleasant View 138kV	Rebuild line with higher-capacity conductor	\$7,327,000	100% (\$7,327,000)
Loudon / Yardley Ridge 138kV	Rebuild line with higher-capacity conductor	\$1,512,000	100% (\$1,512,000)
Indian Lake / Pleasant View 138kV	Rebuild line with higher-capacity conductor	\$11,358,000	100% (\$11,358,000)
Loudon 345-138 kV TX	Replace transformer with higher-rated unit	\$16,041,000	100% (\$16,041,000)
Jet / Poland Rd 138kV	Reconductor line	\$414,000	100% (\$414,000)
	Total allocated to project		\$46,110,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.